

Capstone Adviser Consultation & Defense Preparation Guide

Project: Integrated Dynamic Order, Job Scheduling, and Inventory Management System for Printing Services with Automated Replenishment (Printify)

Event: Adviser Manuscript Review & Consultation • **Target Audience:** Capstone Pair Team

1. Executive Summary: Objective of Consultation

Bukas, magpapacheck kayo ng draft ng inyong Chapter 3 (Methodology) at System Architecture sa inyong capstone adviser. Ang pangunahing layunin ng meeting na ito ay patunayan na **100% nang naaksyunan ang mga pagwawasto mula sa Title Defense** (partikular ang mga mabibigat na komento nina **Ma'am Betg** at **Sir Ryan**), ilatag ang pormal na system title, at ipakita ang natapos na tatlong core diagrams: **Normalized 8-Table ERD (3NF)**, **UML 2.5 Use Case Diagram**, at **11-Class Object-Oriented Class Diagram**.

2. System Title Evolution: Before vs. After

Inalis ang mga kahinaan sa lumang title batay sa mga bilin ng panel:

Stage	Proposed Title	Panel Criticisms	Current Solution
Title Defense 1	<i>"Design and development of a web-based order and production management system..."</i>	Bawal ang "Design and development" dahil automatic objective na ito.	Inalis ang "Design and development". Mas direct sa core system capability.
Title Defense 2	<i>"Web-based order and production management system for Purehandz... with demand forecasting"</i>	1. Alisin ang "Purehandz". 2. Huwag mag-focus lang sa Hardbound. 3. Demand Forecasting needs 3-yr data at formal math model; otherwise ordinary CRUD lang.	1. Generalized SaaS printing model. 2. All print services included. 3. Pinalitan ng Automated Replenishment via BOM & Dynamic ROP telemetry.
Approved Final Title	"Integrated Dynamic Order, Job Scheduling, and Inventory Management System for Printing Services with Automated Replenishment"	Niresolba ang lahat ng komento ng panel habang pinanatili ang scientific merit.	100% compliant sa USM BSIS standards at direct sa tatlong haligi ng system.

3. Title Defense Compliance Matrix (Feedback vs. Resolution)

#	Panelist Feedback / Instruction	Source	Implemented Technical Solution
1	Diagrams need to be specific, ayusin ang pag gamit ng symbols, ensure understanding.	Ma'am Betg	Lahat ng diagrams ay standard UML 2.5: baligtad na pabalik ang <<extend>> arrow sa Use Case, 3-compartment Class diagram na may multiplicities, at 3NF normalized ERD.
2	Hindi pwede na isa lang ang mag work; dapat clear ang modules & functions ng bawat isa.	Ma'am Betg	Malinaw na division of labor: Partner 1 sa Architecture, ERD, Class Diagram, BOM; Partner 2 sa Activity Diagrams, SRS, ISO 25010 Quality Evaluation, at Field Testing.
3	Include Inventory module for production but production module is different to inventory.	Ma'am Betg & Sir Ryan	Ganap na pinaghiwalay ang Production Kanban Hub at Inventory Warehouse Hub. Pinagdurugtong lamang sila ng automated background ServiceBom engine.
4	Huwag mag-focus lang sa hardbound dahil seasonal; include all printing & binding services.	Sir Ryan	Dynamic ShopService catalog + universal OrderItem specifications (JSON) na sumusuporta sa Document Printing, Ring Bind, Loose Leaf, Booklet, at Hardbound.
5	Predictions of raw materials vs orders; mag-accept lang base sa kayang i-cater ng stock.	Sir Ryan	Real-time material feasibility checks gamit ang ServiceBom recipes bago mag-advance ang active jobs sa workshop floor.
6	Alisin ang AI buzzwords; demand forecasting needs clear math/moving average o useless ito.	Sir Ryan	Ipinalit ang Rolling Daily Consumption Velocity (Burn Rate) gamit ang 7-day/30-day moving average para kalkulahin ang dynamic Reorder Point (ROP = Lead Time Demand + Safety Stock).
7	Production management dapat may job scheduling; ilang tao at makina ang gagawa.	Sir Ryan	5-Stage Kanban Pipeline (Queue → Printing → Assembly → QC → Pickup) na may assigned staff operator, machine allocation tag, at stage duration timestamps.
8	GCash recording: upload receipt at i-verify; priority sa full payment at rush orders.	Sir Ryan	Manual payment verification queue para sa GCash/Maya/OTC screenshots bago i-queue ang trabaho; may dynamic rush fee calculation at priority flag.
9	Customer-supplied substrate ("Dala ang Papel") consideration.	Sir Ryan	"Dala ang Papel" toggle sa ordering portal: binabawas ang presyo ng papel sa quote, binabypass ang stock deduction, at naglalagay ng counter verification sa job ticket.
10	Software as a Service (SaaS) multi-tenant framework para sa printing shops.	Sir Ryan	PrintShop multi-tenant root entity na may 1:1 ownership sa User (BusinessOwner) at dynamic per-shop catalog setup.

4. System Architecture & Database Normalization Overhaul

Mula sa lumang 14+ tables na may "table-per-service" anti-pattern, na-normalize ang database sa **8 Unified Core Tables (3NF)**:

- **Users:** Sentral na authentication at Role-Based Access Control (Owner, Staff, Customer) na may Fortify 2FA/passkeys.
- **PrintShops:** Multi-tenant anchor (1:1 sa Owner) na humahawak sa shop settings, address, at branding.
- **ShopServices:** Isang pinag-isang catalog table na may structured settings JSON para sa custom pricing at turnaround formulas.
- **ServiceBoms:** Unified Bill of Materials na nagmamapa ng serbisyo sa raw material consumption (per copy, per page, o duplex per sheet).
- **InventoryItems:** Central inventory tracking para sa raw materials, ready-to-sell, at consumables na may dynamic reorder thresholds.
- **StockMovements:** Immutable audit ledger na nagtatala ng bawat bawas (production), dagdag (restock), o tapon (spoilage) na may direct order traceability.
- **Orders:** Central transactional entity na nag-oorkestra sa 5-stage workshop production pipeline at payment verification.
- **OrderItems:** Universal line-item table na may structured specifications JSON na dynamic na tumatanggap ng anumang print product.

5. Pair Division of Labor & Team Responsibilities

Ito ang pormal na delineation ng inyong magkapares para ipakita kay Adviser na pantay at malinaw ang inyong mga ambag:

Partner 1: System Architect & Backend Lead	Partner 2: Process & Evaluation Lead
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• Technical Architecture & Schema Design: 8-Table 3NF Normalized ERD. • Structural Domain Modeling: 11-Class Object-Oriented UML Class Diagram na may inheritance at composition. • Automated BOM & Telemetry Engine: Dynamic inventory deduction algorithms, duplex sheet math, at burn rate calculations. • Security & Database Implementation: RBAC route middleware, Eloquent models, migrations, at automated test suites.	• Business Process Modeling: Activity Diagrams, end-to-end swimlanes (Customer intake to counter release). • System Requirements Specifications: SRS functional specifications, input-process-output mappings, at UI mockups. • Quality Assurance & ISO 25010 Framework: Evaluation methodology, questionnaire instrument design, at statistical Likert tables. • Empirical Data Gathering: Pakikipag-ugnayan sa tatlong (3) printing establishments para sa field testing at survey execution.
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6. Anticipated Adviser Q&A; Cheatsheet (Ready Answers)

Q1: Bakit pinalitan ang Demand Forecasting sa title? Hindi ba iyon ang approved sa Title Defense?

Sagot: Pinuna po nina Sir Ryan at Ma'am Betg na ang tunay na demand forecasting ay nangangailangan ng 3 taong historical sales data para sa mathematical models tulad ng moving averages, na wala naman ang mga printing shops. Binalaan po kami na magiging basyo o pilit ang AI claim. Kaya iminungkahi po nilang palitan ito ng **Automated Replenishment through Bill of Materials (BOM) and Dynamic Reorder Point (ROP)**. Ito po ay empirical, agad na gumagana, at direktang lumulutas sa problema ng stockouts habang may active rush orders.

Q2: Paano ninyo siniguro na hindi seasonal at hindi lang sa Hardbound thesis nakatali ang system?

Sagot: Inayos po natin ang database architecture gamit ang dynamic ShopServices at OrderItems.specifications JSON pattern. Sinusuportahan na po ng system ang Document Printing (loose leaf, staple, sliding folder, ring bind, booklet) at kahit anong customized print services. Kahit magdagdag ang shop ng bagong serbisyo sa hinaharap, hindi na kailangang magbago o mag-migrate ng database schema.

Q3: Bakit magkahiwalay ang Production Module at Inventory Module?

Sagot: Ayon sa mungkahi ni Ma'am Betg, magkaiba ang concern ng shop floor sa warehouse. Ang **Production Module** ay nakatuon sa time-sensitive workflow—pila ng makina, operator assignment, at 5-stage Kanban queue. Ang **Inventory Module** naman ay nakatuon sa stock balances, unit costs, at spoilage waste. Ang nag-uugnay lamang sa kanila sa background ay ang ServiceBom engine na awtomatikong nagbabawas ng stocks kapag pumasok sa production ang trabaho.

Q4: Paano kung magdala ng sariling papel ang customer ("Dala ang Papel")?

Sagot: Mayroon tayong dedicated toggle para rito. Kapag pinili ito ng customer, awtomatikong binabawas ng pricing formula ang cost ng papel sa quotation, binabypass ng system ang pagbawas ng papel sa shop inventory, at naglalagay ng counter verification alert sa job ticket ng production staff.

7. Checklist para sa Consultation Meeting Bukas

- [] Dala ang printed o digital copy nitong **adviser_consultation_guide.pdf**.
- [] Naka-ready sa laptop ang 3 verified diagrams: **ERD-draft.png**, **use_case_diagram_final.png**, at **class_diagram_draft.png**.
- [] Naka-open ang draft manuscript sa **chapter_3_methodology_draft.md**.
- [] Napag-usapan na ninyong magkapares ang hatian sa pagsagot: Si Partner 1 ang sasagot sa schema at technical engine, at si Partner 2 ang sasagot sa workflows, SRS, at testing.